

6 (long)

(a) State whether each of the following statements is true or false, giving explanations for each of your answers:

(i) When a mass m of a perfect gas is heated at constant pressure such that its temperature rises by ΔT , the increase in its internal energy is given by $\Delta U = mc_p \Delta T$ where c_p is the isobaric specific heat capacity.

(ii) For the low speed flow of an ideal gas through an insulated throttle, the temperatures upstream and downstream of the throttle are approximately equal.

(iii) When a fluid flows steadily through a pipe at high speed, such that frictional effects are significant, the entropy of the fluid necessarily increases in the downstream direction. [9]

(b) Air flows steadily through a horizontal insulated pipe with a uniform cross-sectional area of 0.05 m^2 as illustrated in Fig. 4. At a point A in the pipe the pressure, temperature and velocity of the air are 2.5 bar, 325 K and 400 m s^{-1} respectively. At point B (which may be either upstream or downstream of A) the pressure is 2.0 bar.

(i) Calculate the mass flow in the pipe. [3]

(ii) Calculate the velocity of the air at B. [7]

(iii) Using a thermodynamic approach and explaining your reasoning, determine whether B is upstream or downstream of A. [6]

(iv) Calculate the force exerted by the pipe on the air in the section between A and B and explain how the result may be used to confirm your answer to Part (b)(iii). [5]

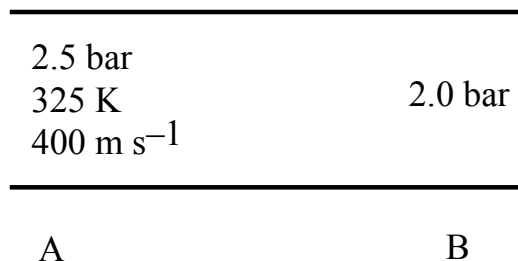


Fig. 4